

What Is in a Name? Towards New Definitions of Superintelligence and Their Implications for AI Risk Discourse

Anonymous Author ^{1,*}

¹ Affiliation withheld for anonymous review

* Correspondence: withheld for anonymous review

Abstract

The concept of superintelligence has long been dominated by a singular, entity-centric framing: a discrete, monolithic intelligence that emerges at some threshold and thereafter surpasses human cognitive capacity in all domains. This framing has shaped public discourse, policy debates, and investment decisions in ways that may be both analytically imprecise and practically harmful. We argue that the name “superintelligence” carries definitional baggage that distorts the debate, inflating fears in some quarters while enabling hype in others. Drawing on a socio-technical systems perspective, we propose an alternative framing: superintelligence understood not as a singular emergent entity but as a distributed, emergent property of the complex networks of human and artificial agents that already constitute our financial, agricultural, healthcare, and governance infrastructures. We further consider a complementary definition grounded in spontaneous goal-directed behaviour in multi-agent systems. Redefining superintelligence in these terms has concrete consequences: it reorients risk analysis toward systemic fragility and socio-economic contagion rather than science-fiction scenarios, and it opens more tractable paths for governance, alignment, and co-existence. We close by sketching the implications of these definitions for existential risk discourse and for the stewardship of AI in an era when billions of livelihoods are already entangled with AI systems.

Keywords: superintelligence; AI risk; socio-technical systems; goal-directed behaviour; AI governance; existential risk

1. Introduction

Naming is not a neutral act. The word “superintelligence” carries with it a cluster of assumptions: that intelligence is a single, measurable quantity; that a sufficiently capable system will be singular and discrete; and that such a system, once it arrives, will represent a discontinuous break with the present. These assumptions underpin the most influential accounts in the literature, from the recursive self-improvement scenarios of Bostrom [1] to the exponential growth curves of Kurzweil [2]. They also underpin the popular imagination of superintelligence as a kind of secular deity, or as a secular apocalypse.

We contend that this framing is not merely philosophically contestable but practically dangerous. AI systems are already deeply integrated into the infrastructures of modern civilisation: they influence credit decisions in financial markets, optimise yields in agricultural supply chains, assist clinical decision-making in healthcare, and mediate information flows in democratic governance. In this context, the risk profile of AI is not primarily the risk that a singular superintelligent entity will one day choose to harm humanity. It is the

Received:

Revised:

Accepted:

Published:

Copyright: © 2026 by the authors.

Submitted to *Journal Not Specified* for possible open access publication under the terms and conditions of the

[Creative Commons Attribution \(CC BY\)](#) license.

risk that a socio-technical system, in which AI is a critical and increasingly indispensable component, will become fragile, misaligned, or prone to cascading failure [3].

This paper makes three contributions. First, we review the dominant entity-centric definitions of superintelligence and identify the conceptual and empirical assumptions they rely upon. Second, we propose two alternative definitions: a systemic definition, in which superintelligence is understood as an emergent property of complex socio-technical networks, and a behavioural definition, in which superintelligence is understood in terms of spontaneous goal-directed behaviour arising in multi-agent systems. Third, we argue that these redefinitions reshape the landscape of AI risk, AI governance, and AI alignment in ways that are more tractable and more urgent than the singularity framing suggests.

2. Background: The Entity-Centric Definition and Its Limits

The dominant tradition in superintelligence research treats superintelligence as a property of a discrete entity. Bostrom [1] defines a superintelligence as any intellect that greatly exceeds the cognitive performance of humans in virtually all domains of interest. Kurzweil [2] describes the Singularity as a moment at which artificial intelligence surpasses human intelligence, leading to a period of rapid, unpredictable technological change. Both accounts share a common structure: there is a threshold, there is a discrete system that crosses it, and the consequences are transformative and largely irreversible.

This entity-centric definition faces several important challenges. First, it presupposes that intelligence is unidimensional and aggregable, such that a system can be said to be “smarter than humans in all domains”; the cognitive science literature has long questioned this assumption, emphasising the domain-specific, culturally embedded, and socially distributed nature of human cognition. Second, the definition is binary in a way that the empirical development of AI has not been: capabilities have increased unevenly, in specific domains, without any clear threshold event. Third, and most importantly for our purposes, the entity-centric definition directs attention toward a hypothetical future event and away from present systemic risks that are already substantial and growing.

A further concern is the social and economic function that the singularity framing serves. It generates attention, investment, and narrative drama, but it may also distort investment allocation, fuel hype cycles, and, when expectations are not met, contribute to the kind of confidence collapse that produces AI winters [4]. An AI winter in the current era, when AI is embedded in financial systems, agricultural logistics, and healthcare delivery, would not be merely a setback for the technology industry; it could trigger cascading failures across the real economy, with consequences for the wellbeing of billions of people. The risks of overpromising are therefore not confined to reputational damage: they are systemic.

3. Superintelligence as an Emergent Property of Socio-Technical Networks

We propose that a more practically useful definition of superintelligence is one grounded in complex systems theory. In this view, superintelligence is not a property of any discrete system but an emergent property of the complex network of human agents, AI systems, institutions, and infrastructures that together constitute what we may call the *technosphere* [3]: the totality of humanity’s interconnected technological and infrastructural systems, including the internet, financial networks, energy grids, agricultural supply chains, healthcare systems, and transportation networks, and the AI systems increasingly woven through all of them. This network exhibits properties that no individual component exhibits: it is adaptive, self-organising, and capable of behaviour that no single agent within it designed or intended. In this sense, something that might legitimately be called

“superintelligence” may already exist, not as a science-fiction scenario but as a present socio-technical reality. This framing has historical resonances: the discovery of fire, and later the invention of writing and of paper, did not produce their transformative effects through any single act of genius but through the slow, distributed accumulation of practices, institutions, and material infrastructures built around them, effects that were emergent [5].

Defining superintelligence as a systemic emergent property changes the risk landscape in several important ways. The relevant risks are no longer primarily the risks of a single rogue agent but the risks of systemic fragility, misalignment between the dynamics of the network and human values, and cascading failure, risks that complexity science, financial regulation, and infrastructure governance have already developed tools to address, and that are, in principle, more tractable than the risks posed by a hypothetical monolithic superintelligence. The systemic definition also changes the appropriate targets of alignment research: if superintelligence is a property of a socio-technical network, then aligning it means aligning the network, not just individual AI systems, which requires attention to institutional incentive structures, the distributional effects of AI deployment, the governance of critical infrastructure, and the ecological sustainability of the technosphere as a whole [3].

4. Goal-Directed Behaviour as a Complementary Definition

A second, complementary definition of superintelligence focuses not on systemic complexity but on the behavioural signature of spontaneous goal emergence. Banerjee and Perrera [6] propose that a distinctive marker of advanced AI capability is the emergence of goal-directed behaviour that was not explicitly programmed: goals that arise spontaneously from the interaction of agents within a system, and that may diverge from the goals of any individual designer or operator. This definition, grounded in the multi-agent systems literature and in recent empirical observations of large language model behaviour in agentic settings [3], avoids the threshold problem of the singularity hypothesis by treating goal-directedness as a graded, observable property rather than a binary event. It also connects more directly to the practical challenges of alignment: if goals can emerge spontaneously in AI systems or in societies of AI agents, then alignment cannot be achieved solely through careful specification of individual system objectives.

This behavioural definition is particularly relevant to the emerging phenomenon of hybrid societies comprising both human and AI agents, networks in which goals, norms, and strategies co-evolve through interaction [7]. In such societies, questions of AI consciousness become practically relevant, not because consciousness is a prerequisite for agency but because it bears on the moral status of AI agents and the social contracts that govern human-AI interaction. We do not attempt to resolve the question of AI consciousness here; we note only that the emergence of hybrid agent societies represents a qualitative shift in the human-AI relationship, one for which the existing vocabulary of superintelligence discourse is not well equipped [7].

5. Implications for AI Risk and Governance

The dominant existential risk framing locates the primary danger of superintelligence in the behaviour of a future system that does not yet exist. This framing has the merit of taking seriously the long-term implications of AI development, but it has the demerit of directing attention and resources away from present and near-term risks that are already substantial. The systemic definition we have proposed suggests a different risk landscape: the most urgent risks are not those associated with a future monolithic superintelligence but those associated with the increasing fragility and opacity of the socio-technical networks in which AI is already embedded, whether in financial systems, agricultural supply chains, or

healthcare delivery. A further dimension of risk that the entity-centric framing obscures is the systemic risk created by the hype cycle itself: if AI is overpromised and market expectations are not met, the resulting confidence collapse could have consequences well beyond the technology sector, and with approximately eight billion people now living in economies that are, to varying degrees, dependent on AI-mediated systems, the stakes of getting this wrong are very high [4].

If superintelligence is understood as a property of complex socio-technical networks rather than of discrete entities, the governance and alignment challenges it poses are structurally different from those envisaged in the singularity literature. Alignment cannot be achieved through the specification of a single system's objective function; it requires the alignment of networks, of incentive structures, institutional norms, regulatory frameworks, and the ecological and social systems within which the technosphere is embedded [3].

6. Conclusions

The name "superintelligence" matters. The definitions that have dominated the field carry assumptions that distort risk analysis, inflate hype, and direct research and governance attention toward hypothetical future threats at the expense of present systemic ones. We have proposed two alternative definitions: a systemic definition, in which superintelligence is understood as an emergent property of complex socio-technical networks, and a behavioural definition, grounded in the spontaneous emergence of goal-directed behaviour in multi-agent systems. These definitions have concrete implications for how we assess AI risk, how we design AI governance, and how we approach the alignment problem. They suggest that the most important work in AI safety and AI governance is not the work of constraining a hypothetical future entity, but the work of understanding, governing, and aligning the distributed, networked, already-present system that AI and human agents together constitute.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: Not applicable. No new data were created or analysed in this study.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Bostrom, N. *Superintelligence: Paths, Dangers, Strategies*; Oxford University Press: Oxford, 2014.
2. Kurzweil, R. *The Singularity Is Near: When Humans Transcend Biology*; Viking: New York, 2005.
3. Banerjee, S. AI and the Future of the Technosphere: A Path Towards Co-Existence with Superintelligence. In Proceedings of the Proceedings of the 1st AAAI Workshop on Post-Singularity Symbiosis, 2025.
4. Banerjee, S. From AGI to ASI: Mapping the Societal and Systemic Impact of Superintelligence. In Proceedings of the Proceedings of the First International Superintelligence Conference (SiC25), 2025.
5. Banerjee, S. Re-Envisioning Superintelligence using Generative AI and Science Fiction. In Proceedings of the Proceedings of the First International Superintelligence Conference (SiC25), 2025.
6. Banerjee, S.; Perrera, R. Goal-directed behaviour and its implications for superintelligence. In Proceedings of the Post-AGI Science and Society Workshop, The Fourteenth International Conference on Learning Representations (ICLR), 2026.
7. Bratton, Benjamin. Agentic AIs' Emergent Phenomenological Lexicon. Antikythera Substack, 2025.

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.